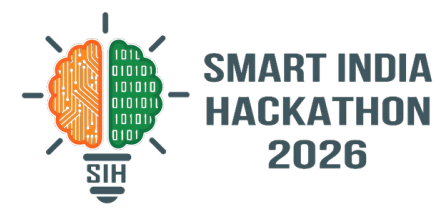
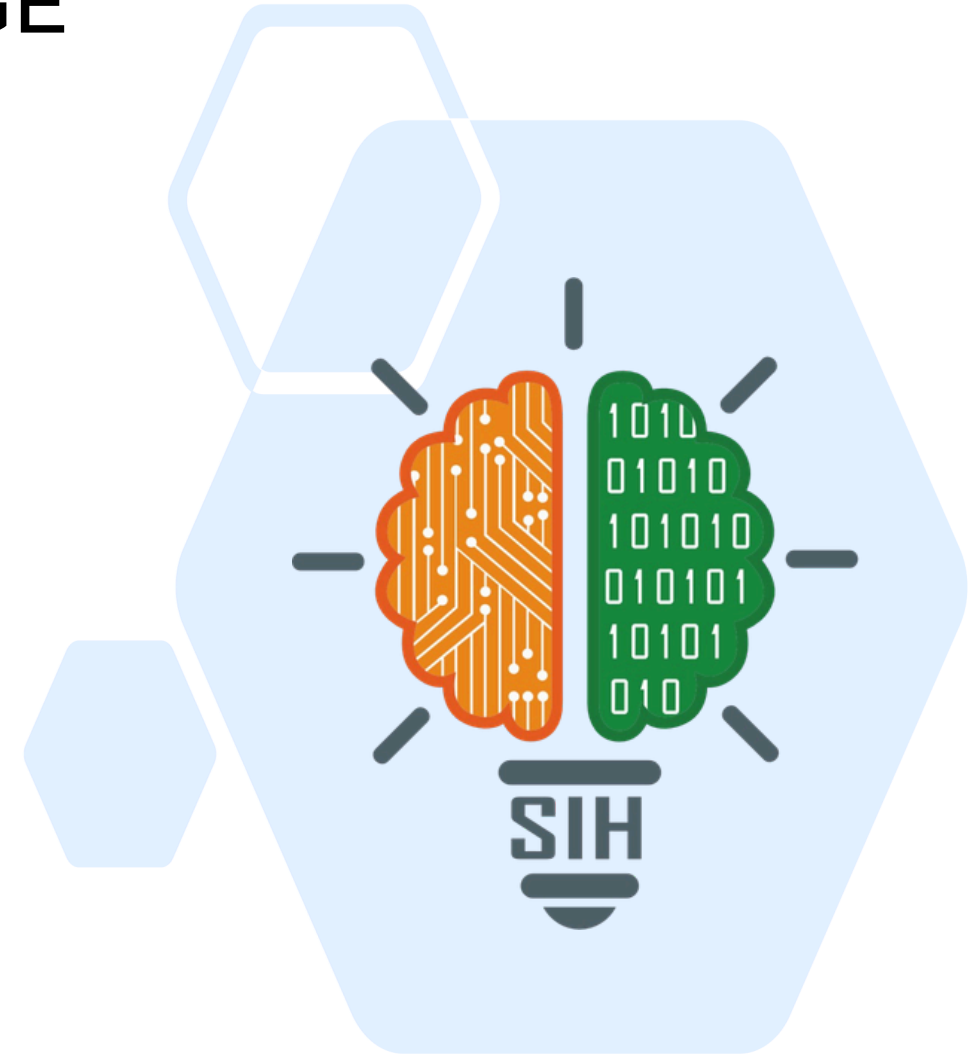


SMART INDIA HACKATHON 2026



TITLE PAGE

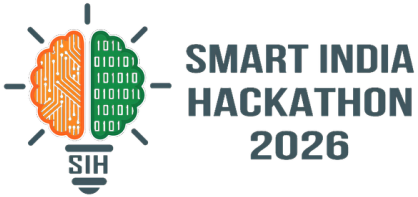
- **Problem Statement ID** – 26169
- **Problem Statement Title** – AI-Based Virtual Camera Tracking
System for Coarse Alignment of Mobile FSOC Terminals
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – —
- **Team Name** – ZeroDrift





IDEA TITLE

ZeroDrift

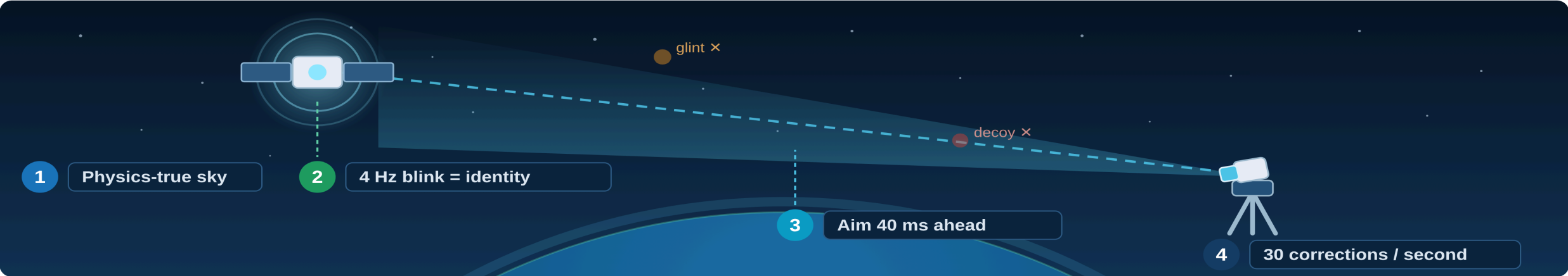


PROPOSED SOLUTION (DESCRIBE YOUR IDEA/SOLUTION/PROTOTYPE)

The eyes and neck of a laser-communication terminal

ISRO · PS 26169 — every figure measured

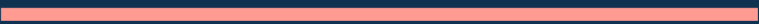
DETAILED EXPLANATION OF THE PROPOSED SOLUTION



INNOVATION AND UNIQUENESS OF THE SOLUTION

Brightness is not identity — modulation is.

Star / glint
brighter, steady



REJECTED

Our beacon
dimmer, 4 Hz



LOCKED

HOW IT ADDRESSES THE PROBLEM



1.27s
TO ACQUIRE

98.3%
LOCK HELD

0/64
WRONG LOCKS

TECHNICAL APPROACH

CORE



Python 3.11 · NumPy · OpenCV

INTERFACE



PySide6 console · pyqtgraph live plots

ORBITS · AI



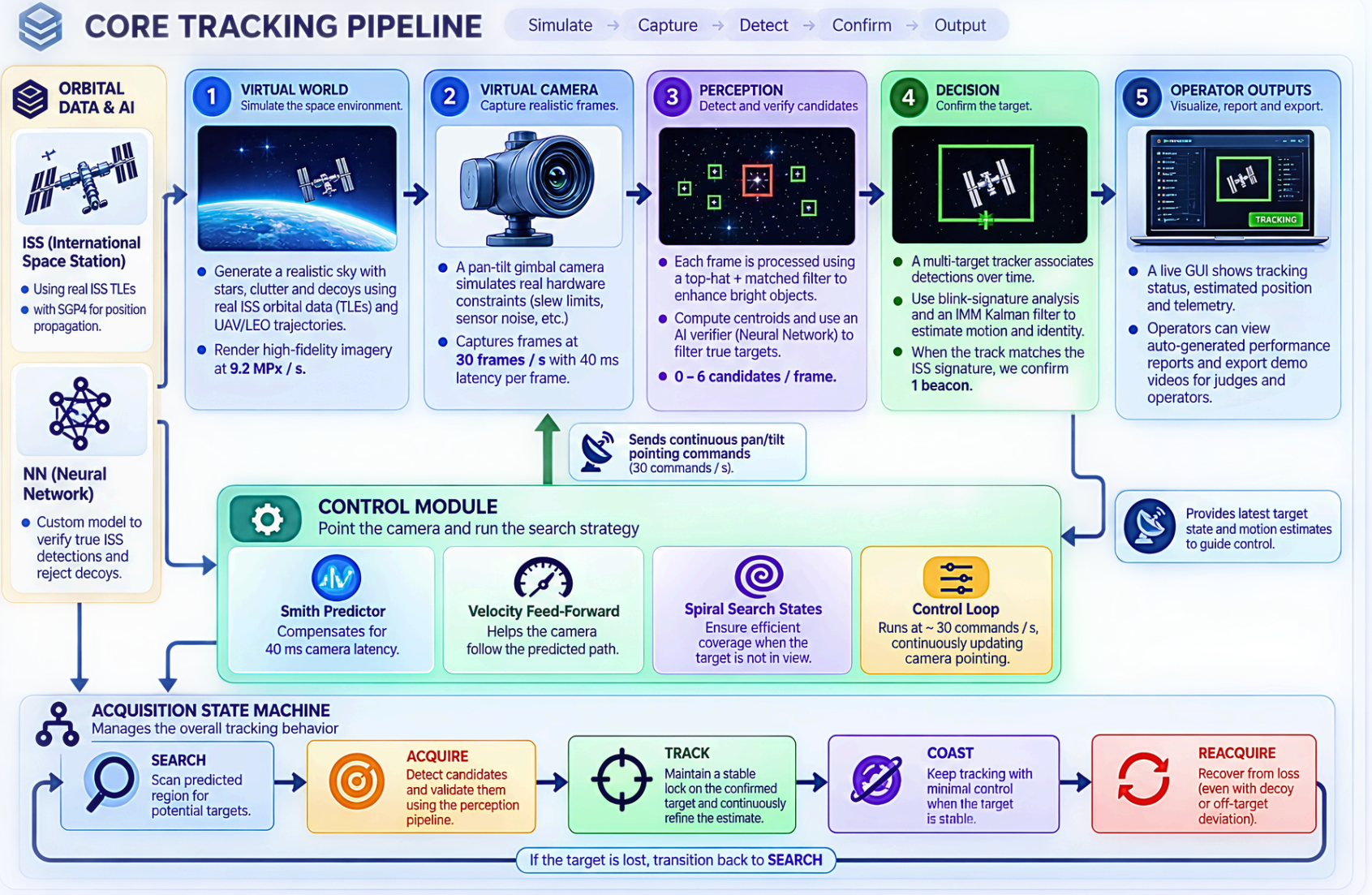
real ISS TLEs · Neural Network

QUALITY



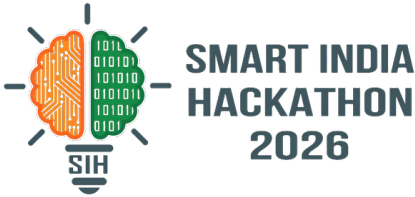
pytest suite · GitHub Actions

SHIPPING





FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA

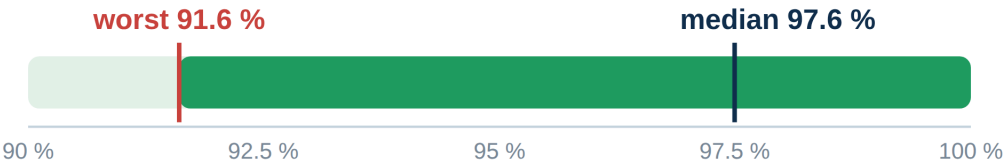
64 randomised runs

dim beacons · heavy turbulence · vibration · moving decoys

Time to acquire the beacon



Lock retention across the pass



Scored against ground truth the tracker never sees. Every run replayable from its logged seed.

CHALLENGES & RISKS → STRATEGIES TO OVERCOME THEM



Heavy turbulence

→ Coast the fade, then re-spiral

1.27 s
94.3 % lock



Decoys & sun-glint

→ 4 Hz blink gate + AI verifier

0 / 64
AUC 0.957



40 ms mount latency

→ Two-path Smith predictor

4.2 μ rad lag
at 0.75 °/s



Sharp UAV manoeuvres

→ Dual-model IMM estimator

re-weighted
every frame

100% 73 / 73
tests pass

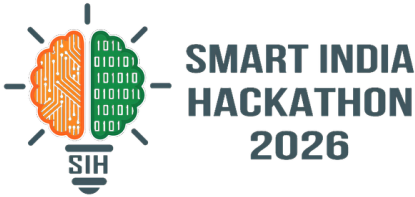
100% 64 / 64
runs acquired

0 wrong locks
decoys never fooled it

8–21 ms of 33 ms
no GPU · no cloud · no dataset



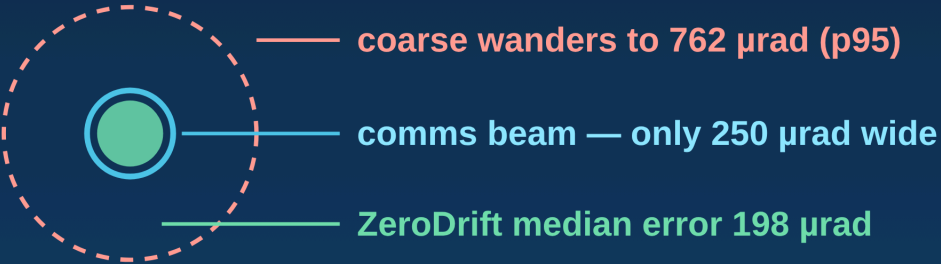
IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

Does the optical link actually close?

Fraction of a real ISS pass with the modelled link closed.



Coarse alignment alone



14.7 %

Coarse + modelled fine stage



99.3 %

6.8× more link time

Tracked ISS pass · 250 μ rad beam · 3.1 dB margin · zero outages.
Measured, not asserted.

two-stage architecture, justified by measurement

What it replaces

An optical test bench

₹ lakhs

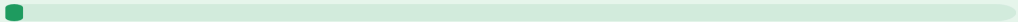
Cameras, gimbals, optics, lab time. One team at a time.



ZeroDrift

₹ 0

A laptop you already own. One command. A whole class at once.



no GPU

no cloud

no dataset

no consumables

open source

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

STRATEGIC

Sovereign FSOC pointing capability, developed in India.

ECONOMIC

Lakhs of rupees of hardware → zero recurring cost.

RESEARCH & EDUCATION

Any lab or student can test PAT algorithms today.

SCALABLE & GREEN

Satellite, UAV, ground, maritime — one pure-software engine.

Who it serves:

ISRO / IN-SPACe optical-communication teams

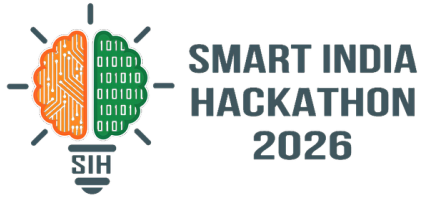
DRDO

IITs & engineering colleges

India's space-tech start-ups



RESEARCH AND REFERENCES



Research foundations

Kaymak et al., “A Survey on Acquisition, Tracking and Pointing Mechanisms for Mobile FSO,” IEEE Comm. Surveys & Tutorials, 2018. [\[link\]](#)

Kaushal & Kaddoum, “Optical Communication in Space: Challenges and Mitigation Techniques,” IEEE Comm. Surveys & Tutorials, 2017. [\[link\]](#)

Blackman & Popoli, Design and Analysis of Modern Tracking Systems — IMM filtering, CFAR detection, track management. [\[link\]](#)

Bar-Shalom et al., Estimation with Applications to Tracking and Navigation — Kalman/IMM theory. [\[link\]](#)

Smith, O.J.M., “Closer Control of Loops with Dead Time” — the Smith predictor, 1957. [\[link\]](#)

Vallado & Crawford, “SGP4 Orbit Determination” — AIAA 2008; real ISS TLE propagation. [\[link\]](#)

Andrews & Phillips, Laser Beam Propagation through Random Media — turbulence & scintillation models. [\[link\]](#)



Our work —open and verifiable

Repository: github.com/pranshukesavmishra/SIH-2026

Live demo (replay console): zerodrift-fsoc-pat.netlify.app

Inside: full source (5,000+ lines, documented) · 73 automated tests · 6 validated scenarios · Monte-Carlo campaign logs · technical report · user manual · demo video.

Reproducibility: every figure in this deck regenerates from one command; every random draw is logged and replayable.

Why ZeroDrift wins on this problem

- ✓ All six mandatory deliverables already exist today
- ✓ Judged numbers, honestly measured against hidden ground truth
- ✓ AI where it earns its place — benchmarked against classical methods

Compared with existing approaches			
Feature	ZeroDrift	MATLAB sim	Traditional
Zero hardware cost No special equipment required	✓	✓	✗
Real-time closed loop + live console See it run. Live.	✓	✗	✓
Decoy & blink-signature discrimination Robust against spoofing	✓	✗	✗
Open source • one-command reproducible Run with a single command	✓	✗	✗